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Evaluation of surface temperature and pressure derived from MERRA-2 and ERA5 reanalysis datasets and their applications in hourly GNSS precipitable water vapor retrieval over China

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ABSTRACT

Temperature and pressure play key roles in Global Navigation Satellite System (GNSS) precipitable water vapor (PWV) retrieval. The National Aeronautics and Space Administration (NASA) and European Center for Medium-Range Weather Forecasts (ECMWF) have released their latest reanalysis product: the modern-era retrospective analysis for research and applications, version 2 (MERRA-2) and the fifth-generation ECMWF reanalysis (ERA5), respectively. Based on the reanalysis data, we evaluate and analyze the accuracy of the surface temperature and pressure products in China using the measured temperature and pressure data from 609 ground meteorological stations in 2017 as reference values. Then the accuracy of the two datasets and their performances in estimating GNSS PWV are analyzed. The PWV derived from the pressure and temperature products of ERA5 and MERRA-2 has high accuracy. The annual average biases of pressure and temperature for ERA5 are −0.07 hPa and 0.45 K, with the root mean square error (RMSE) of 0.95 hPa and 2.04 K, respectively. The annual average biases of pressure and temperature for MERRA-2 are −0.01 hPa and 0.38 K, with the RMSE of 1.08 hPa and 2.66 K, respectively. The accuracy of ERA5 is slightly higher than that of MERRA-2. The two reanalysis data show negative biases in most regions of China, with the highest to lowest accuracy in the following order: the south, north, northwest, and Tibet Plateau. Comparing the GNSS PWV calculated using MERRA-2 (GNSS MERRA-2 PWV) and ERA5 (GNSS ERA5 PWV) with the radiosonde-derived PWV from 48 co-located GNSS stations and the measured PWV of the co-location radiosonde stations, it is found that the accuracy of GNSS ERA5 PWV is better than that of GNSS MERRA-2 PWV. These results show the different applicability of surface temperature and pressure products from MERRA-2 and ERA5 data, indicating that both have important applications in meteorological research and GNSS water vapor monitoring in China.

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1. Introduction

As the most active component in the atmosphere, water vapor is crucial to numerous atmospheric processes, including global energy and water cycling, global warming, and climate change [1–4]. The precipitable water vapor (PWV) is the total atmospheric water vapor contained in a vertical air unit column from the Earth's surface to the top of the atmosphere, which reflects the distribution and variation of atmospheric water vapor in the troposphere [5–7]. The tropospheric signal delay caused by atmospheric water vapor is the major error source and limited factor of several data collection systems, including Global Navigation Satellite System (GNSS),

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Interferometric Synthetic Aperture Radar (InSAR), Very Long Baseline Interferometry (VLBI) and radar survey [8,9]. Therefore, an accurate estimation of atmospheric water vapor is beneficial to the improvement of positioning accuracy. Many studies have researched the impact of the crucial parameters of the troposphere, such as the atmospheric weighted mean temperature (T_m), zenith total delay (ZTD), zenith wet delay (ZWD), and zenith hydrostatic delay (ZHD) [10–13]. Surface temperature and pressure are two key parameters in GNSS PWV retrieval. The traditional method of PWV retrieval by ground-based GNSS relies on the measured meteorological data of stations. However, most GNSS stations are not equipped with corresponding meteorological sensors, which limits the application of a large amount of GNSS data, the research of water vapor variation, and the monitoring with high accuracy and temporal-spatial resolution. Atmospheric reanalysis data with a wide spatio-temporal range provide various meteorological parameters worldwide including temperature and pressure, which can compensate for the flaw of spatio-temporal distribution of meteorological observation data. Therefore, many scholars regard it as a supplementary data source for GNSS PWV [14]. Moreover, atmospheric reanalysis is widely used in all types of meteorological research [15].

Currently, PWV derived from reanalysis data have been researched in many studies [16–19]. The National Aeronautics and Space Administration (NASA) and the European Centre for Medium-Range Weather Forecasts (ECMWF) have released their latest reanalysis data: the modern-era retrospective analysis for research and applications, version 2 (MERRA-2) and the fifth-generation ECMWF reanalysis (ERA5), respectively. There are several researches on the accuracy of temperature and pressure products of reanalysis data. Based on the measured meteorological data, we analyzed the accuracy of the temperature and pressure products of ERA5 and ERA-Interim in China and the error of GNSS PWV derived from ERA5 and ERA-Interim dataset, which was caused by meteorological parameters [20]. In the other research, the accuracy of ERA5 was preliminarily evaluated using radiosonde data in China and then used to estimate GNSS PWV with a time resolution of 1 h. The results demonstrated that the ERA5 temperature and pressure product had high accuracy in China [21].

The above studies have investigated the temperature and pressure products in different regions of China and their application in water vapor research. However, there is a lack of comparative studies of high-precision reanalysis data from different institutions. ERA5 and MERRA-2 provide hourly surface temperature and pressure products. At present, although the relevant literature has preliminarily carried out GNSS PWV retrieval research using ERA5 temperature and pressure products, few studies evaluate the applicability of MEERA-2 reanalysis data in estimating GNSS PWV over China. Therefore, we perform a comparative analysis of the surface temperature and pressure accuracy from MERRA-2 and ERA5 using measured data from 609 surface meteorological stations and explore the accuracy of GNSS PWV retrieved using these two temperature and pressure products, which provides a reference for the subsequent use of reanalysis data and GNSS water vapor retrieval in China.

2. Data and methodology

2.1. Data resource

The recent high-resolution global atmospheric reanalysis data launched by NASA is called MERRA-2. It has a horizontal spatial resolution of 0.5° × 0.625° (longitude × latitude) and a temporal resolution of 1 h for surface data, which can be obtained from https://goldsmr4.gesdisc.eosdis.nasa.gov/data/MERRA-2 [22]. ERA5 is a new global atmospheric reanalysis data provided by the ECMWF, with a horizontal spatial resolution of 0.25° × 0.25° (longitude × latitude) and a temporal resolution of 1 h (https://cds.climate.copernicus.eu/). The measured surface temperature and pressure data were obtained from the China Meteorological Data Center Service (http://data.cma.cn), with a temporal resolution of 1 h. PWV data from radiosonde (RS) stations are available from the University of Wyoming website and were used to evaluate the calculated PWV (http://weather.uwyo.edu/upperair/sounding.html). ZTD product of the Crustal Movement Observation Network of China (CMONC)was downloaded from the GNSS data product service platform (http://www.cgps.ac.cn) of the China Earthquake Administration. Based on the principle of distance within 40 km, 48 RS stations co-located with GNSS stations were selected to reduce the effects of distance and elevation. According to geographical location and climate type, this study divided China into four regions: the south, north, northwest, and Tibet Plateau. Fig. 1 shows the distribution of the regional meteorological (MET) stations, radiosonde stations, GNSS co-location stations and geographical division in China, in which the blue, yellow, red, and green represent meteorological stations located in the southern, northern, northwestern, and Tibet Plateau regions, respectively.

2.2. Methodology

Because the ground stations generally do not coincide with the grid points of the reanalysis data, the height datum of these two datasets is different. GNSS and MET data use the geodetic height system and geoid height system, respectively, while the geopotential height system is used for reanalysis data. The difference between geoid height system and geopotential height system is small and negligible, but the difference between geodetic height and geopotential height could have a significant influence on the height correction of the meteorological parameters. So the different height systems must be unified. In this work, the official Earth Gravitational Model 2008 (EGM2008) was adopted to unify the height datum [23,24]. Owing to the elevation difference between the ground station and its four nearby grid points, the vertical interpolation correction of temperature and pressure data was carried out using the temperature and pressure lapse rate model; and its interpolation formula is presented in Eqs. (1)–(3) [25]. According to the method, the data of the nearest four grid points at the same height of the site were calculated using the temperature and pressure products of two reanalysis data, the bilinear interpolation method was used for horizontal interpolation of the temperature and pressure data, and these data of the reanalysis data at the site were obtained.

$$T = T_0 - \gamma(h_1 - h_2)\partial \tag{1}$$

$$P = P_0\left[1 - \frac{\gamma(h_1 - h_2)}{T_0}\right]^{\frac{g\cdot M}{R\cdot\gamma}} \tag{2}$$

$$g = 9.8063\cdot\left\{1 - 10^{-7}\frac{h_1 - h_2}{2}\left[1 - 0.0026373\cdot\cos(2\varphi) + 5.9\cdot10^{-6}\cdot\cos^2(2\varphi)\right]\right\} \tag{3}$$

Where T and T_0 (K) represent the corresponding temperature values at the height of h_1 and h_2 (m), respectively; P and P_0 (hPa) represent the corresponding pressure values at the height of h_1 and h_2, respectively; φ is vertical lapse rate, generally taken as the mean constant (−0.0065 K/m), M stands for the molar mass of dry air, which is typically 0.0289644 kg/(mol·K); R is the ideal gas

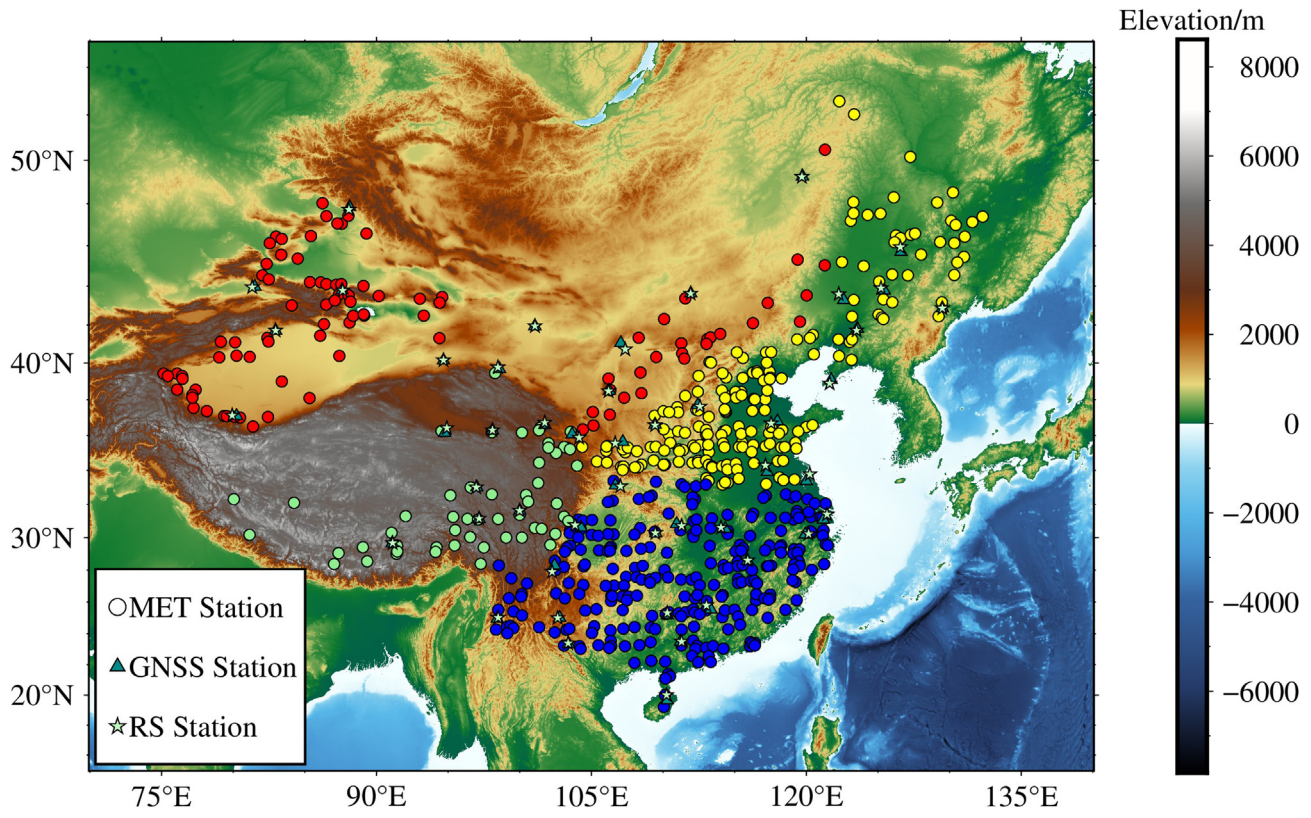


Fig. 1. Distribution of ground meteorological stations, radiosonde stations, GNSS stations and the four geographical regions in China.

constant and usually takes as 8.31432 $N \cdot m/(mol \cdot K)$; g stands for the gravitational coefficient.

Based on the ZTD from the GNSS observation data in China and the ZHD at the GNSS station calculated by the Saastamoinen model [26]:

$$ZHD = \frac{2.2767 \times P_s}{1 - 0.00266 \cos 2\varphi - 0.0028 \times h_0} \tag{4}$$

where P_s, φ, h_0 represent the GNSS station's surface pressure (in hPa), the latitude (in radians) of the GNSS station, and the ellipsoidal height (in km) of the station, respectively.

Then, GNSS PWV could be calculated according to Eqs. (5) and (6) [27].

$$ZWD = ZTD - ZHD \tag{5}$$

$$PWV = \frac{10^6}{\rho R_v \left[(k_3/T_m + k_2')\right]} \cdot ZWD \tag{6}$$

where $\rho = 1 \times 10^3$ kg/m^3 is the liquid water density, k_2', k_3 are the atmospheric physical parameters, R_v is water gas constant, the values of k_2', k_3 and R_v are 22.13 ± 2.20 K/hPa, (3.739 ± 0.012) × 10^5 K^2/hPa, and 461.495 J· kg^{-1}·K^{-1}, respectively. T_m is the atmospheric weighted average temperature. Based on the single-factor regression model of regional weighted average temperature in China [28], this paper divides China into four regions, and calculates T_m through T_s at GNSS stations. T_s is the surface temperature. Table 1 shows the four T_m models.

Fig. 2 shows the GNSS PWV retrieval process. After calculating the PWV at each GNSS station, the PWV from the radiosonde station has to be vertically interpolated to the same height as the GNSS PWV when comparing these two PWV data. The Chinese regional PWV vertical correction model that considers time-varying factors was adopted in the PWV vertical interpolation [29].

$$PWV_{h1} = PWV_{h2} \cdot \exp(\beta(h_1 - h_2)) \tag{7}$$

Where PWV_{h1} represents PWV values corresponding to the heights h_1, PWV_{h2} represents PWV values corresponding to the heights h_2, and β is the PWV lapse rate. The annual mean, annual cycle and semi-annual cycle model coefficients of the lapse rate are shown in detail in Ref. [29].

3. Results

3.1. Evaluation of temperature and pressure from MERRA-2 and ERA5 reanalysis data

Surface temperature and pressure data provided by the ground meteorological stations were obtained from meteorological sensors with high accuracy and reliability. Therefore, we used surface temperature and pressure products at ground MET stations in China during 2017 to evaluate and analyze the accuracy of ERA5 and MERRA-2. First, the temperature and pressure data from the ground meteorological station were calculated by the interpolation

Table1
T_m models of the four geographical regions in China.

Classification	Model
Southern China	$T_m = 90.10 + 0.650\ T_s$
Northern China	$T_m = 86.87 + 0.658\ T_s$
Northwestern China	$T_m = 140.63 + 0.467\ T_s$
Tibet Plateau	$T_m = 144.09 + 0.452\ T_s$

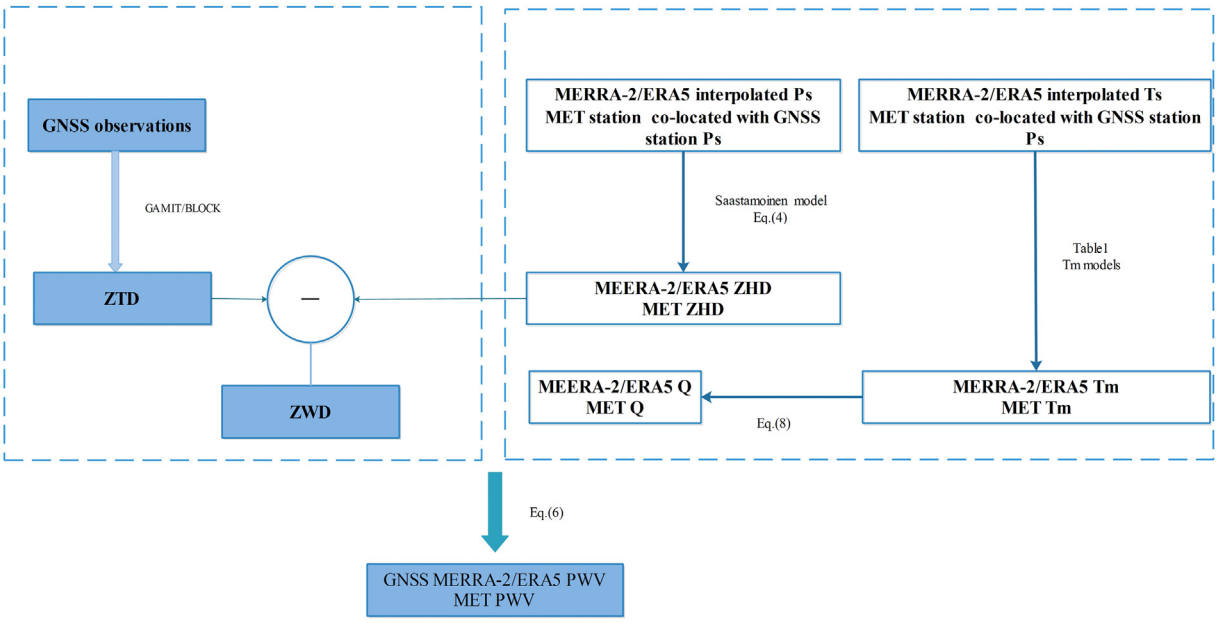


Fig. 2. GNSS PWV retrieval process.

of MERRA-2 and ERA5 reanalysis data, and then compared with the ground meteorological station data. The accuracy index consists of the bias and the root mean square error (RMSE).

To better analyze the temperature and pressure products of two reanalysis data in China, the country was divided into four regions. The annual average bias and RMSE distribution of temperature and pressure products from two reanalysis data in China are shown in Fig. 3 and Fig. 4, and the bias statistics of each region are shown in Table 2.

According to Figs. 3 and 4, ERA5 shows an obvious negative bias throughout the majority of China, especially in the Tibet Plateau. The temperature and pressure products of MERRA-2 also indicate a negative bias in the southern, northern, and Tibet Plateau regions and a positive bias in the northwestern region, especially in Xinjiang, where the biases of temperature and pressure products from two types of reanalysis data are approximately −2 to 2 K and −1 to1 hPa, respectively. ERA5 has a large bias and RMSE in the Tibet Plateau, and MERRA-2 shows a large bias and RMSE in northwestern China. The reason is that the undulating terrain and complex climate of Tibet Plateau and Xinjiang result in a dramatic or subtle variation in temperature and pressure data, and few GNSS stations are another factor. Overall, the two reanalysis data of temperature and pressure products are slightly lower than the measured temperature and pressure data in the whole of China.

From Table 2, the temperature products provided by two reanalysis data have good precision in the whole of China, and their annual biases are 0.38 and 0.45 K, respectively. However, in terms of RMSE, ERA5 performs better than MERRA-2, and the former ranges from 0.93 to 6.04 K. The average temperature of ERA5 is 2.04 K, while that of MERRA-2 ranges from 1.37 to 9.44 K, with an annual bias of 2.66 K. Therefore, the accuracy of ERA5 in China is slightly higher than that of MERRA-2. As for pressure assessment, the annual bias of MERRA-2 and ERA5 reanalysis data ranges from −6.98 to 7.70 and −7.02 to 8.15 hPa, respectively, with an average value of −0.01 and −0.07 hPa, respectively. The mean bias of two reanalysis data shows significant changes, but it is also small in the whole of China. The average RMSEs of two reanalysis data are 1.08 and 0.95 hPa, respectively, which are almost the same. The results show that the temperature and pressure products provided by the ERA5 and MERRA-2 reanalysis data are highly precise and reliable in China. For this study area, the precision of EAR5 temperature and pressure products from high to low is: the southern, northern, northwestern, and Tibet Plateau regions. And the precision of MERRA-2 temperature product ranks from high to low as follows: southern China, northern China, northwestern China, and Tibet Plateau. The precision of MERRA2 pressure product from high to low is northern China, southern China, northwestern China, and Tibet Plateau. In general, the temperature and pressure products of ERA5 were slightly more accurate than those of MERRA-2 in all regions of China, which might be because ERA5 shows a better spatial resolution than MERRA-2. These reanalysis data have high precision in all regions of China, even in the northwest and Tibet Plateau regions, which the precision values are relatively the worst. The temperature RMSE was within 4.04 K, and the pressure RMSE was within 1.54 hPa, which can satisfy the needs of fundamental meteorological research and water vapor retrieval.

3.2. Analysis of suitability of ERA5 and MERRA-2 in hourly GNSS PWV retrieval

To evaluate the influence of temperature and pressure product of two reanalysis data on GNSS PWV, the errors of PWV retrieval were analyzed, which mainly include conversion factor derived from temperature and ZHD derived from pressure. Eqs. (8) and (9) are used to calculate the conversion factor and errors, respectively. The errors caused by surface pressure on ZHD can be calculated according to Eq. (10) [30].

$$Q = \frac{10^6}{\rho R_v \left[\left(k_3 / T_m + k_2' \right) \right]} \tag{8}$$

$$\sigma_Q = 10^{-6} \rho R_v \sqrt{\left(\frac{\sigma_{k_3}}{T_m} \right)^2 + \sigma_{k_2'}^2 + \left(k_3 \frac{\sigma_{T_m}}{T_m^2} \right)^2} \tag{9}$$

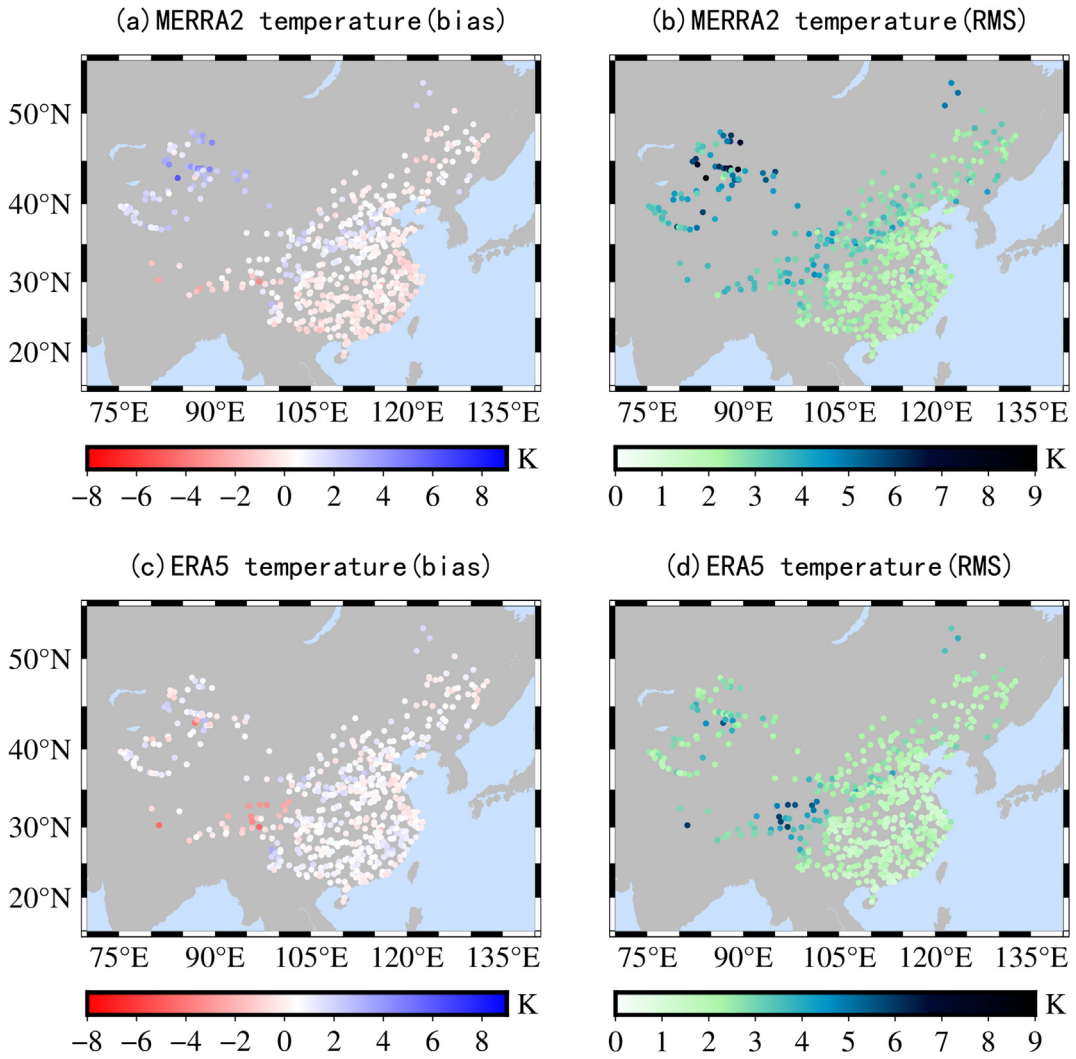


Fig. 3. Distribution of mean bias and RMSE of temperature from MERRA-2 and ERA5 in China.

$$\sigma_{\mathrm{ZHD}} = \sqrt{\left(\frac{2.2767\sigma_{P_s}}{f(\lambda,H)}\right)^2 + \left(\frac{P_s\sigma_c}{f(\lambda,H)Q}\right)^2} \tag{10}$$

$$f(\lambda,H) = \left(1 - 2.66\times10^{-3}\cos(2\varphi) - 2.8\times10^{-7}H\right) \tag{11}$$

Where σ_Q σ_{k_3} $\sigma_{k_2'}$ and σ_{T_m} are the errors of conversion factor Q, k_3 k_2' and T_m , respectively. P_s is the pressure at the meteorological station, σ_{P_s} stands for the error of pressure, λ is latitude of the station, H is geodetic height of the station, and σ_c is taken as 0.0024.

The errors of PWV can be estimated according to Eq. (12)

$$\sigma_{\mathrm{PWV}} = \sqrt{\left(\frac{\sigma_{\mathrm{ZTD}}}{Q}\right)^2 + \left[\frac{2.2767\sigma_{P_s}}{f(\lambda,H)Q}\right]^2 + \left[\frac{P_s\sigma_c}{f(\lambda,H)Q}\right]^2 + \left(\mathrm{PWV}\cdot\frac{\sigma_Q}{Q}\right)} \tag{12}$$

Where σ_{ZTD} σ_{p_s}, and σ_Q represent the errors of ZTD, pressure, and conversion factor Q, respectively.

The average temperature error of two reanalysis data is about 2.5 K, and the relative error caused by the conversion factor is less than 1%. The ZHD error caused by pressure is about 2 mm per 1 h Pa. Compared with the measured pressure at meteorological stations, the errors of two reanalysis data are within 2 hPa at most stations, and ZHD error caused by them is also less than 4 nm. According to the error propagation law, the effect of temperature and pressure products on GNSS PWV error using two reanalysis data is less than 1 mm.

To analyze the influence of temperature and pressure products for MERRA-2 and ERA5 on GNSS PWV retrieval in China, we take the GNSS PWV derived from the measured temperature and pressure data of 48 meteorological stations co-located with GNSS stations as the reference value (GNSS MET PWV). Fig. 5 and Fig. 6 show the annual mean biases of PWV derived from the MERRA-2 (GNSS MERRA-2 PWV) and ERA5 data (GNSS ERA5 PWV).

The annual mean biases of GNSS ERA5 PWV and GNSS MERRA-2 PWV are 0.54 and 0.48 mm, respectively, which means they both have high precision in China (Figs. 5 and 6). The annual mean RMSEs are 1.04 and 1.2 mm, respectively. Therefore, the GNSS ERA5 PWV is slightly more consistent with GNSS MET PWV than GNSS MERRA-2 PWV at most sites. To analyze the accuracy of temperature and pressure products in retrieving GNSS PWV, the GNSS MERRA-2 PWV and GNSS ERA5 PWV were evaluated and compared with PWV data from radiosonde stations co-located with a GNSS station in 2017 as the reference value. The results are presented in Table 3 and Fig. 7.

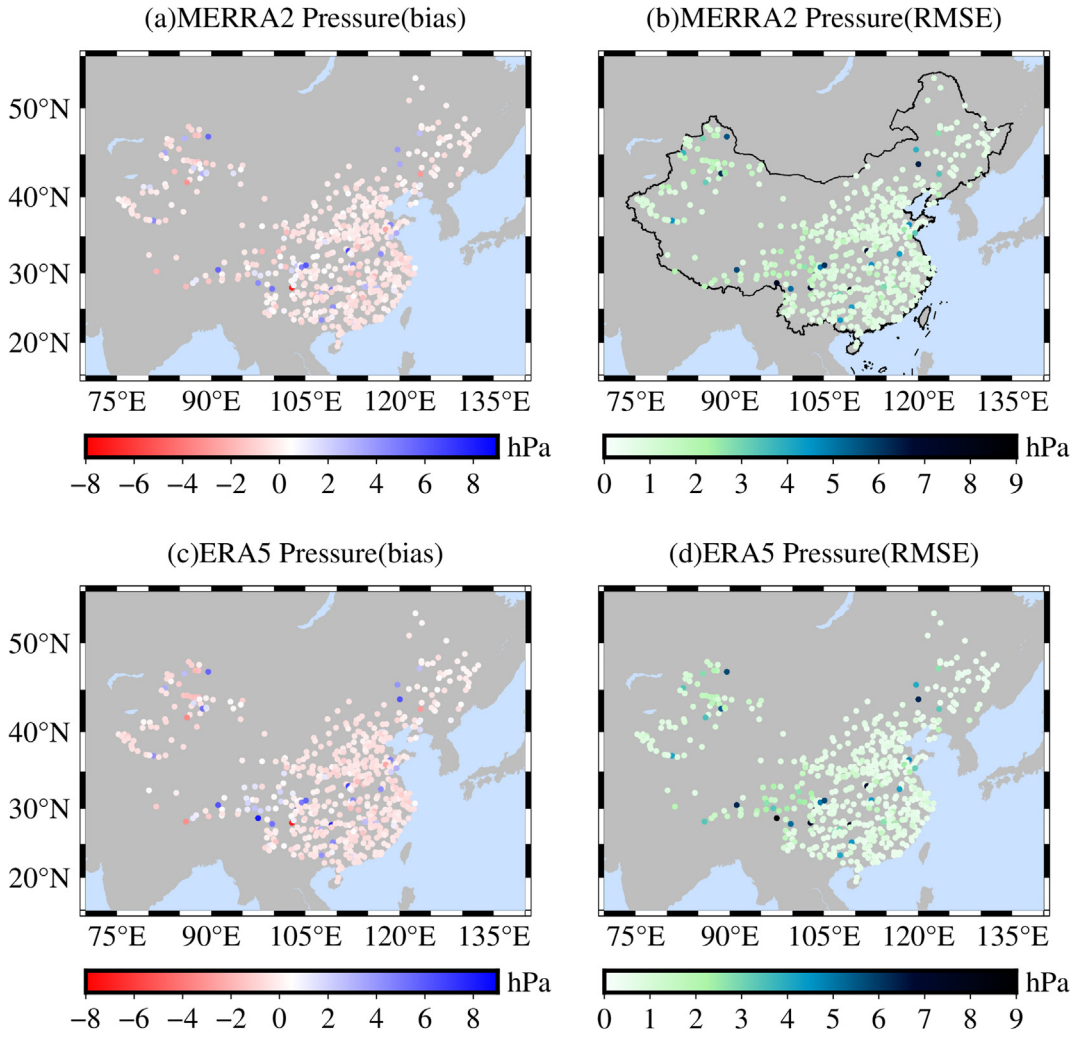


Fig. 4. Distribution of mean bias and RMSE of pressure from MERRA-2 and ERA5 in China.

Table 2
Statistical results of mean bias and RMSE of temperature and pressure from MERRA-2 and ERA5 reanalysis data in China.

region	Temperature bias (K)		Temperature RMSE (K)		Pressure bias (hPa)		Pressure RMSE (hPa)	
	ERA5	MERRA-2	ERA5	MERRA-2	ERA5	MERRA-2	ERA5	MERRA-2
Southern China	−0.09	−0.55	1.60	2.04	−0.09	−0.02	0.91	1.08
Northern China	−0.43	−0.49	1.93	2.52	−0.10	−0.21	0.70	0.82
Northwestern China	0.56	1.61	2.47	3.38	0.10	0.15	1.19	1.35
Tibet Plateau	−0.37	−0.46	3.37	4.04	−0.61	0.13	1.54	1.45
China	0.45	0.38	2.04	2.66	−0.07	−0.01	0.95	1.08

The GNSS MERRA-2 PWV and GNSS ERA5 PWV exhibit a negative bias across most regions of China when compared to the radiosonde PWV (Fig. 6). The GNSS ERA5 PWV and GNSS MERRA-2 PWV show a large bias in southern China compared to other regions. The RMSEs of the GNSS ERA5 PWV and GNSS MERRA-2 PWV in most parts of China are less than 2 mm, but the RMSEs of some stations in southern China are approximately 4 mm. In general, the accuracy of the GNSS ERA5 PWV and GNSS MERRA-2 PWV in the whole of China is relatively high, but the accuracy of them in southern China is obviously lower than that in other regions. The reason may be the annual rainfall in southern China is relatively high and the variation in water vapor is more severe than in other regions, thus causing a large error. According to Table 3, compared with the PWV values of the radiosonde, the annual bias fluctuation of GNSS ERA5 PWV in China is less than that of GNSS MERRA-2 PWV, which ranges from −2.46 to 0.79 mm, while GNSS MERRA-2 PWV ranges from −2.67 to 1.02 mm. The mean biases of GNSS ERA5 PWV and GNSS MERRA-2 PWV are −0.43 and −0.57 mm, respectively, showing significant negative bias. The annual RMSE of GNSS ERA5 PWV in China ranges from 0.68 to 3.61 mm, while that of GNSS MERRA-2 PWV ranges from 0.51 to 3.82 mm, and the average RMSE is 2.06 and 2.16 mm, respectively. Therefore, the accuracy of the

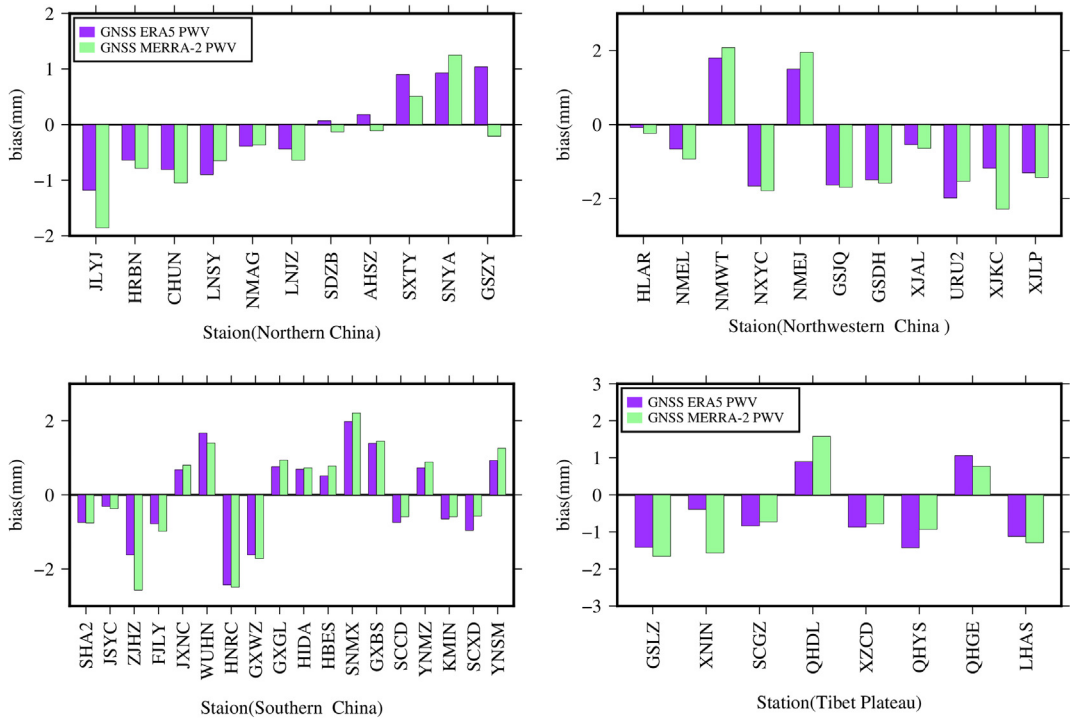


Fig. 5. Annual mean bias of PWV derived from ERA5 and MERRA-2 at GNSS stations in China.

GNSS MET PWV retrieved by surface temperature and pressure product from ERA5 is slightly higher than that from MERRA-2 over China.

The regions with high to low accuracy of the GNSS ERA5 PWV and GNSS MERRA-2 PWV are: the north, Tibet Plateau, northwest, and south. The accuracy of the GNSS ERA5 PWV is better than that of the GNSS MERRA-2 PWV in south and north China. However, the accuracy of the GNSS ERA5 PWV is lower than that of the GNSS MERRA-2 PWV in the northwest and Tibet Plateau, which may be since there are few co-located stations in those regions. The error

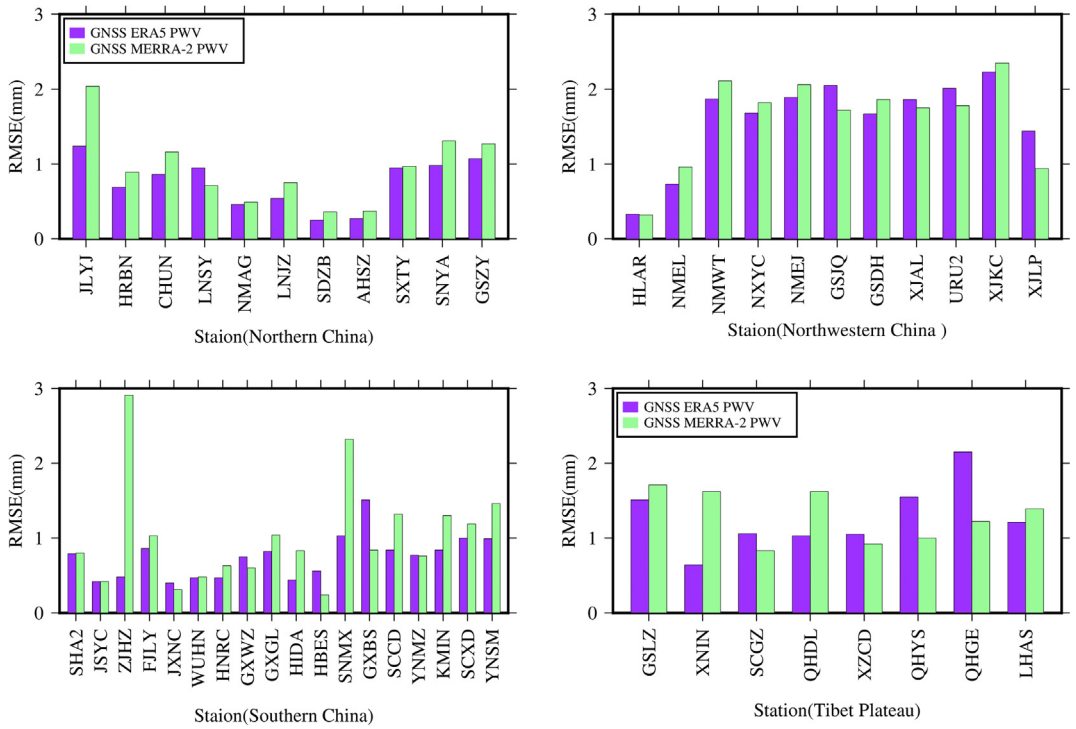


Fig. 6. Annual mean RMSE of PWV derived from ERA5 and MERRA-2 at GNSS stations in China.

Table 3
Statistical results of mean bias and RMSE of GNSS MERRA-2 PWV and GNSS ERA5 PWV with respect to radiosonde data in different regions of China.

Region	Bias (mm)		RMSE (mm)	
	ERA5	MERRA-2	ERA5	MERRA-2
Southern China	−0.49	−0.53	2.77	2.95
Northern China	0.09	0.14	1.72	2.07
Northwestern China	−0.66	−1.12	1.79	1.65
Tibet plateau	−0.68	−0.80	1.42	1.38
China	−0.43	−0.57	2.06	2.16

of a single station has a significant impact on the evaluation of the overall accuracy of the region, whereas the overall accuracy of the GNSS ERA5 PWV in China is higher than that of the GNSS MERRA-2 PWV. The main reason may be that the ERA5 shows higher horizontal resolution than MERRA-2, which causes a higher accuracy of interpolated meteorological parameters than MERRA-2. However, the bias of the two reanalysis data is both within 2 mm, and the RMSE values are within 3 mm. This shows high accuracy in terms of GNSS PWV retrieval. Therefore, we conclude that good accuracy can be obtained by applying surface temperature and pressure derived from ERA5 and MERRA-2 in calculating GNSS PWV, which can provide powerful data sources for high-precision GNSS PWV data acquisition and water vapor change research in China.

4. Conclusion and discussion

In this study, we take measured atmospheric temperature and pressure data from 609 ground meteorological stations in 2017 as reference values, which is aimed to analyze the accuracy and applicability of temperature and pressure product from MERRA-2 and ERA5 in China. Meanwhile, the accuracy of its application in GNSS PWV retrieval is analyzed. Compared with the measured temperature and pressure of 609 ground meteorological stations, the average annual biases of temperature and pressure derived from ERA5 and MERRA-2 data are 0.45 K and −0.07 hPa, and 0.38 K and −0.01 hPa, respectively; the average annual RMSE are 2.04 K and 0.95 hPa, and 2.66 K and 1.08 hPa, respectively. Both reanalysis data have high accuracy and applicability in China, while the accuracy of ERA5 is higher than that of MERRA-2. The temperature and pressure products of both reanalysis data show a negative bias in most regions of China.

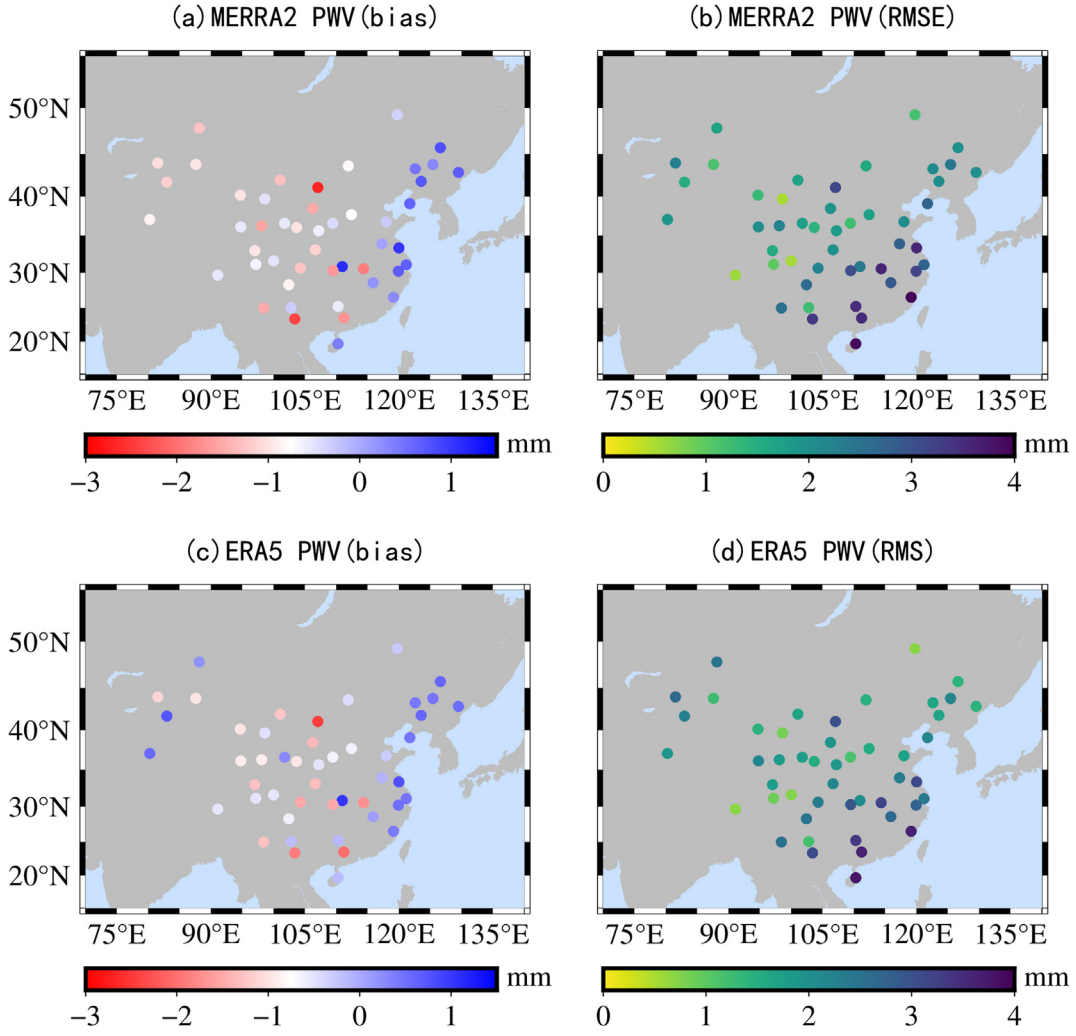


Fig. 7. Distribution of mean bias and RMSE of GNSS MERRA-2 PWV and GNSS ERA5 PWV tested by radiosonde data in China.

Compared with the PWV values measured by radiosonde stations, the average annual RMSE of GNSS PWV derived from ERA5 and MERRA-2 surface temperature and pressure in China are 2.06 and 2.16 mm, respectively. Overall, the accuracy of GNSS ERA5 PWV is higher than that of GNSS MERRA-2 PWV.

The PWV derived from MERRA-2 and ERA5 with GNSS PWV is of high application, although extreme weather and other climate studies still need to be addressed in more detail. Due to the climate, precipitation patterns, complicated geography, and the lack of GNSS stations in some locations, these elements should be taken into account in the integrated multi-source PWV research in China.

In conclusion, the surface temperature and pressure accuracy derived from ERA5 and its application to GNSS PWV retrieval results are slightly higher than those of MERRA-2. The surface temperature and pressure products of two reanalysis data applied in GNSS PWV retrieval have good accuracy and can be used as powerful supplementary data resources for GNSS PWV retrieval and regional meteorological research in China.

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Conflicts of interest

The authors declare that there is no conflicts of interest.

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